

C14 S2

18. Three Entrances A study was conducted to aid in the remodeling of an office building that contains three entrances. The choice of entrance was recorded for a sample of 200 persons who entered the building. Do the data in the table indicate that there is a difference in preference for the three entrances? Find a 95% confidence interval for the proportion of persons favoring entrance 1.

Entrance	1	2	3
Number Entering	83	61	56

$$H_0: p_1 = p_2 = p_3 = \frac{1}{3}, \quad H_a: p_i \neq \frac{1}{3}, \quad i = 1, 2, 3$$

$$\alpha = 0.05$$

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{k-1}$$

$$E_i = (83 + 61 + 56) \frac{1}{3} = 66.67$$

$$\chi^2 = \frac{(83 - 66.67)^2 + (61 - 66.67)^2 + (56 - 66.67)^2}{66.67} = 6.19$$

95% CI

$$p = \hat{p} \pm z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\hat{p} = \frac{83}{200} = 0.415 \quad z_{\frac{\alpha}{2}} = z_{0.025} = 1.96$$

$$p = 0.415 \pm 1.96 \sqrt{\frac{0.415(1-0.415)}{200}} = (0.347, 0.483)$$

$$RR: \chi^2_{0.05, 3-1} = 5.99147$$

$$\{ \chi^2 \mid \chi^2 > 5.99147 \}$$

$$6.19 > 5.99147 \rightarrow \text{Reject } H_0$$

There is sufficient evidence to say that there is a difference in preference for the three entrances.